AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2011

Mathematics Paper II

Time allowed: 2 hours 20 minutes Marks 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name ad school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer ALL ELEVEN questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

(ATTEMPT EITHER PART a OR PART b OF Q.1.)

Q.1. (Total 5 Marks)

a. Find the highest common factor (H.C.F) of (x^2-4), (x^3+8) and $(x+2)^2$ by factorization method.

b. Simplify $\dfrac{27xy^3-3x^3y}{(3y-x)^2} \div \dfrac{3xy}{3y-x}$.

Q.2. (Total 3 Marks)

If $\dfrac{8-x}{(x-1)(x-2)} = \dfrac{A}{x-1} + \dfrac{B}{x-2}$, then find the values of *A* and *B*.

(ATTEMPT EITHER PART a OR PART b OF Q.3.)

Q.3. (Total 5 Marks)

a.

i. Find the value of x for $\frac{3-x}{2}+\frac{6-x}{3}=0$. (3 Marks)

ii. Find the solution set of $a-x>a$, where $x \in N$. (2 Marks)

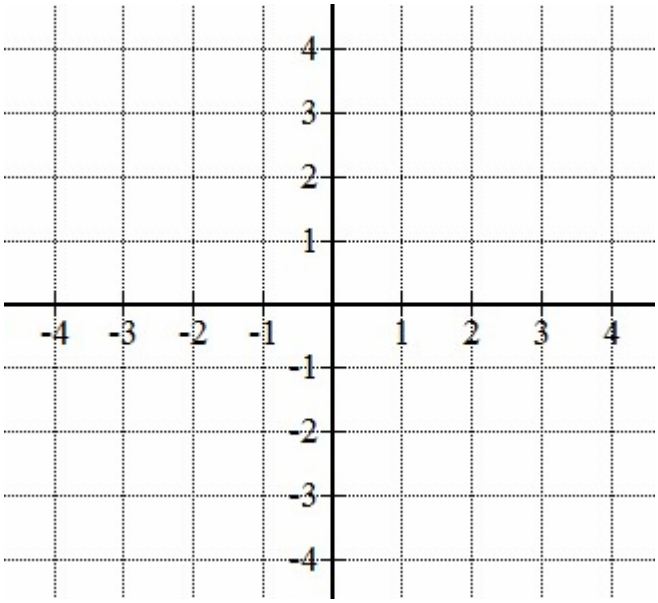
b.

i. There are 96 CDs placed by Salman and Sobia in a CD rack. One fifth of the rack is still vacant. Find the total capacity of the CD rack. (3 Marks)

ii. Find the solution set of $|x-1|=1$. (2 Marks)

Q.4. (Total 5 Marks)

i. Locate (0, 3), (3, 0), (2, 1) and (1, 2) on the following graph paper. Are these points forming a triangle? Justify your answer. (3 Marks)



ii. Complete the following table for the linear equation $3x - 5y = 23$. (2 Marks)

x		5
y	5	

Q.5. (Total 4 Marks)

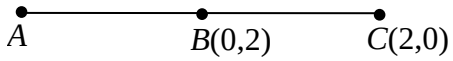
Find the values of n for $n^2 - n + \frac{2}{9} = 0$ by factorization method.

(ATTEMPT EITHER PART a OR b OF Q.6.)

Q.6. (Total 3 Marks)

a. Find the shortest distance between points $A(2,-2)$ and $B(-2,2)$.

b. If $A(x, y), B(0,2)$ and $C(2,0)$ are three collinear points such that $mAB = mBC$,then find the coordinates of point A.

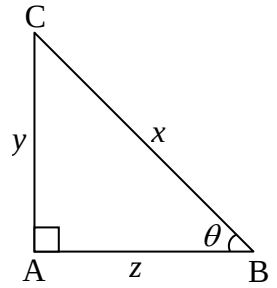


(ATTEMPT EITHER PART a OR b OF Q.7.)

Q.7. (Total 4 Marks)

a. Prove that $\left(\sec^2\theta + \operatorname{cosec}^2\theta\right) \div \left(\frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta}\right) = \sec\theta \operatorname{cosec}\theta$. (4 Marks)

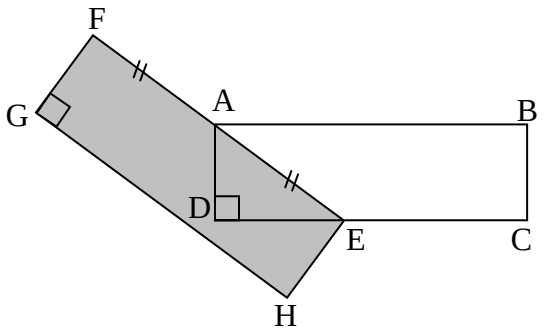
b. Consider the following right angled triangle *ABC* and prove that $1 + \tan^2\theta = \sec^2\theta$. (4 Marks)



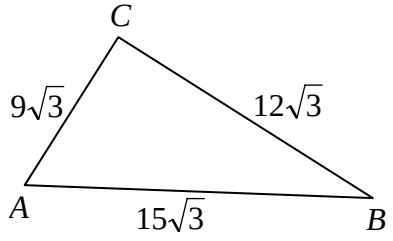
(ATTEMPT EITHER PART a OR b OF Q.8.)

Q.8. (Total 4 Marks)

a. If two congruent rectangles *ABCD* and *EFGH* are overlapped as shown in the figure, find the length of *DE*, when $mAB = 20\text{cm}$, $GF = 9\text{cm}$ and $FA = AE$. (4 Marks)

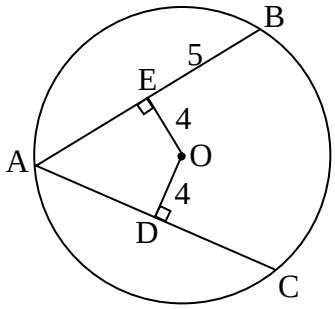


b. Prove that ΔABC is a right angled triangle. Identify the right angle of the triangle. (4 Marks)



Q.9. (Total 4 Marks)

In the following figure, O is the centre of the circle.

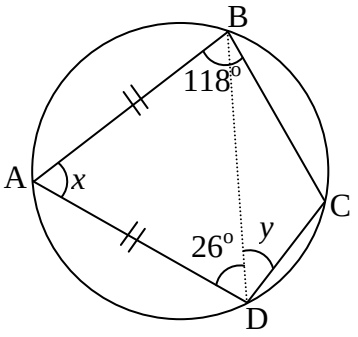


i. Find the measure of *DC* and give a reason for your answer. (2 Marks)

ii. Find the measure of *AB* and give a reason for your answer. (2 Marks)

Q.10. (Total 4 Marks)

For the given figure, find the angles *x* and *y* if $\overline{AB} = \overline{AD}$.



Q.11. (Total 4 Marks)

Construct a circle of radius 2.5cm and inscribe a square in the circle.

END OF PAPER

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